# IGCSE 0607 Practice Questions — E7.5 — Combined Questions & Markscheme

## Non-right-angled triangles

**Total Questions:** 27 — **Total Marks:** 187 Marks

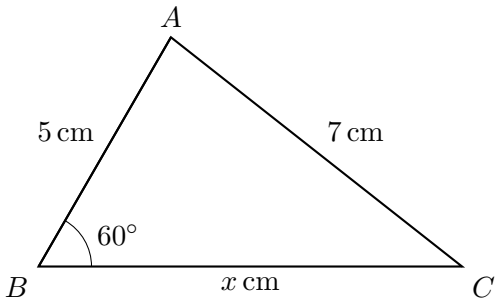
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- **E7.5** Non-right-angled triangles
  → **Paper 2 (Non-Calculator)** *(1 Q)* · **Paper 4 (Calculator)** *(26 Qs)*


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**1** In triangle $ABC$, $AB = 5\,\text{cm}$, $BC = x\,\text{cm}$ and $AC = 7\,\text{cm}$.
Angle $ABC = 60°$.




NOT TO SCALE

**(a)** Use the cosine rule to show that $x^2 - 5x - 24 = 0$.

**[3]**

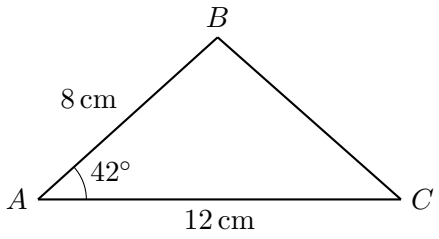
**(b)** Solve the equation $x^2 - 5x - 24 = 0$ to find the length of $BC$.

$BC =$ ..... cm **[2]**

**(c)** Calculate the exact area of triangle $ABC$.
Give your answer in the form $a\sqrt{b}$, where $a$ and $b$ are integers.

Area = ..... $cm^2$ **[2]**

**2** The diagram shows triangle $ABC$.
$AB = 8\,\text{cm}$, $AC = 12\,\text{cm}$ and angle $BAC = 42°$.



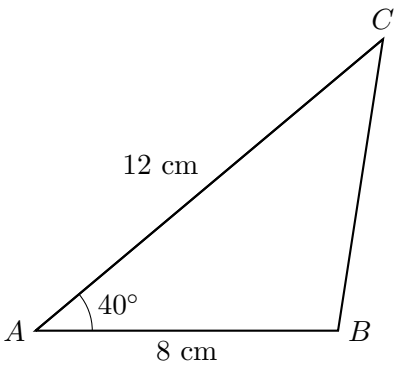

**(a)** Calculate the area of triangle $ABC$.

..... $cm^2$ **[2]**

**(b)** Calculate the length of $BC$.

$BC =$ ..... cm **[3]**

**3** The diagram shows a triangle $ABC$.



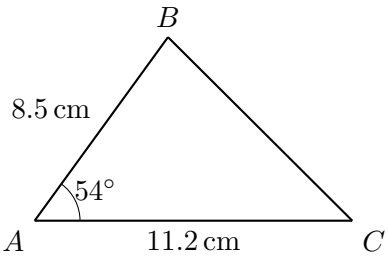

**(a)** Calculate the area of triangle $ABC$.

..... $cm^2$ **[2]**

**(b)** Calculate the length of $BC$.

..... cm **[3]**

**4** The diagram shows triangle $ABC$.



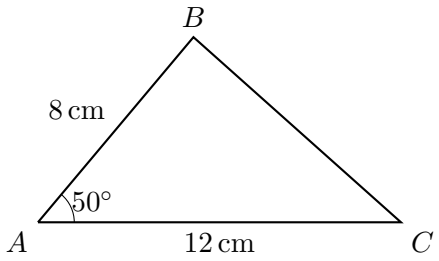

**(a)** Calculate the area of triangle $ABC$.

..... $cm^2$ **[2]**

**(b)** Calculate the length of $BC$.

..... cm **[3]**

**5** A triangle $ABC$ is shown below.




NOT TO SCALE

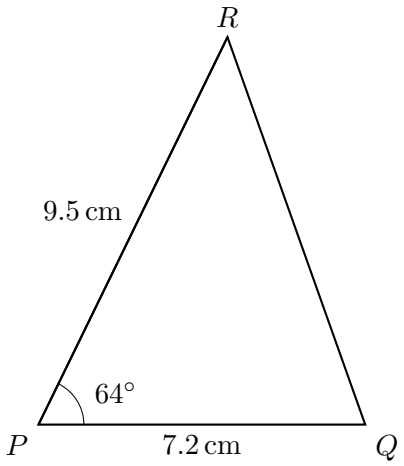
**(a)** Calculate the area of triangle $ABC$.

..... $cm^2$ **[2]**

**(b)** Calculate the length of $BC$.

..... $cm$ **[3]**

**6** In triangle $PQR$, $PQ = 7.2\,\text{cm}$, $PR = 9.5\,\text{cm}$ and angle $QPR = 64°$.




NOT TO SCALE

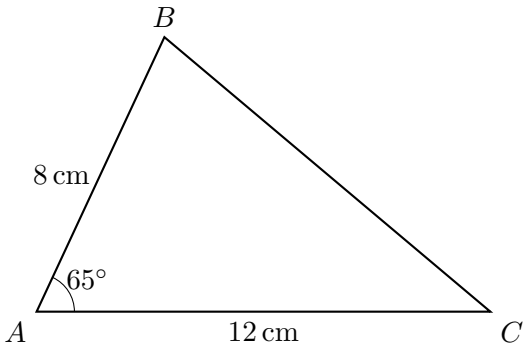
**(a)** Calculate the area of triangle $PQR$.

Area = ..... $cm^2$ **[2]**

**(b)** Calculate the length of $QR$.

$QR$ = ..... cm **[3]**

**7** The diagram shows a triangle $ABC$ with $AB = 8\,\text{cm}$, $AC = 12\,\text{cm}$ and angle $BAC = 65°$.




NOT TO SCALE

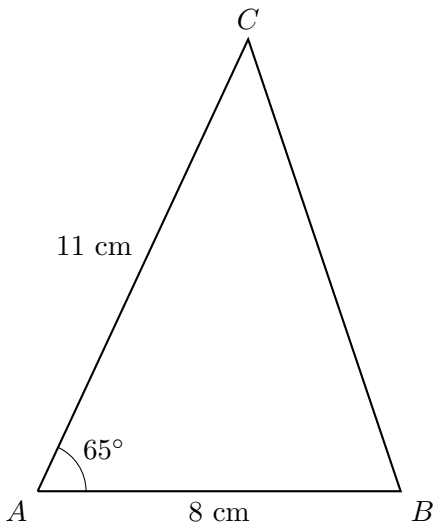
**(a)** Calculate the length of $BC$.

$BC =$ ..... cm **[3]**

**(b)** Calculate the area of triangle $ABC$.

Area = ..... $cm^2$ **[2]**

**8** The diagram shows a triangle $ABC$.
$AB = 8$ cm, $AC = 11$ cm and angle $BAC = 65°$.



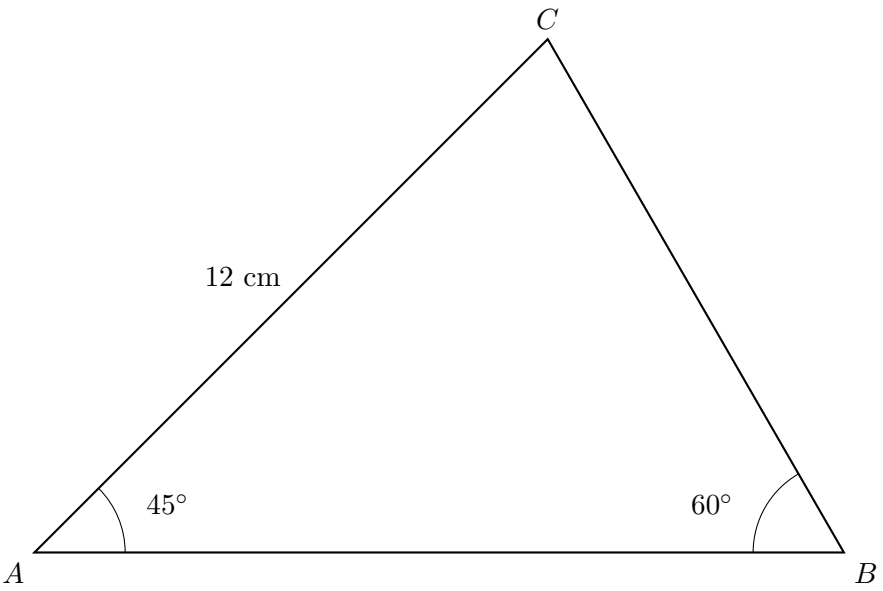

**(a)** Calculate the area of triangle $ABC$.

Area = ..... $cm^2$ **[2]**

**(b)** Calculate the length of $BC$.

$BC$ = ..... cm **[3]**

**9** In triangle $ABC$, $AC = 12$ cm, angle $BAC = 45°$ and angle $ABC = 60°$.



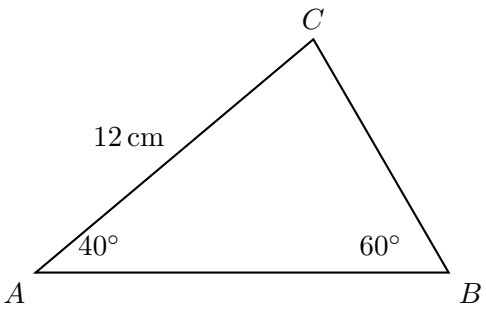

**(a)** Calculate the length of $BC$.

..... cm **[3]**

**(b)** Calculate the area of triangle $ABC$.

..... $cm^2$ **[3]**

**10** The diagram shows triangle $ABC$.
$AC = 12$ cm, angle $BAC = 40°$ and angle $ABC = 60°$.



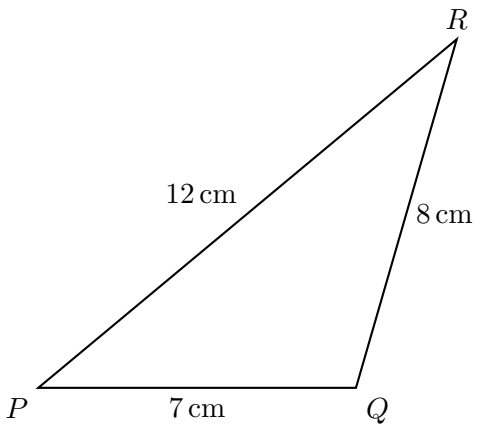

**(a)** Calculate the length of $BC$.

$BC =$ ..... cm **[3]**

**(b)** Calculate the area of triangle $ABC$.

Area = ..... $cm^2$ **[3]**

**11** The diagram shows triangle $PQR$.
$PQ = 7$ cm, $QR = 8$ cm and $PR = 12$ cm.



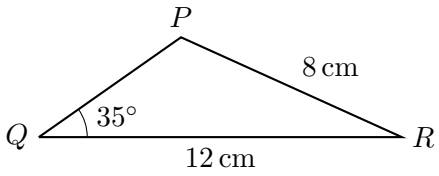

**(a)** Calculate the largest angle in triangle $PQR$.

..... ° **[3]**

**(b)** Calculate the area of triangle $PQR$.

Area = ..... $cm^2$ **[2]**

**12** In triangle $PQR$, $PR = 8$ cm, $QR = 12$ cm and angle $PQR = 35°$.
Angle $QPR$ is an obtuse angle.




NOT TO SCALE

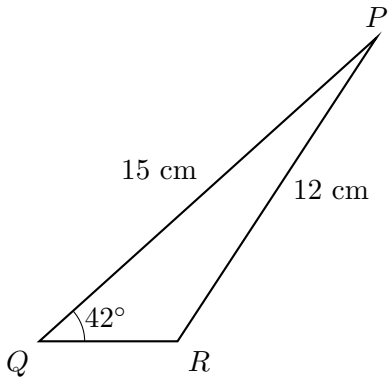
**(a)** Calculate the size of angle $QPR$.

..... ° **[4]**

**(b)** Calculate the area of triangle $PQR$.

..... $cm^2$ **[2]**

**13** The diagram shows a triangle $PQR$.
The length of $PQ$ is 15 cm and the length of $PR$ is 12 cm.
Angle $PQR = 42°$.
Angle $PRQ$ is obtuse.



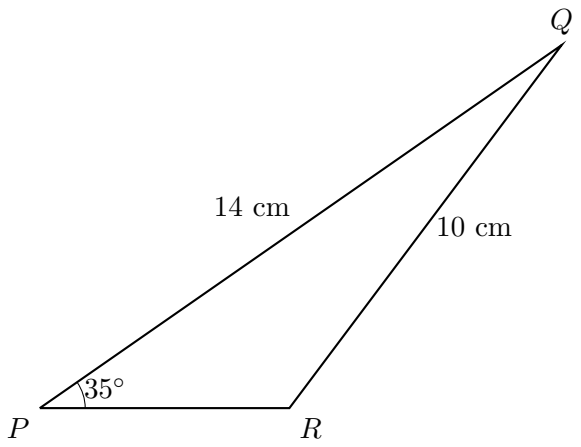

**(a)** Calculate angle $PRQ$.

..... ° **[3]**

**(b)** Calculate the length of $QR$.

..... cm **[3]**

**14** The diagram shows triangle $PQR$.
$PQ = 14$ cm, $QR = 10$ cm and angle $QPR = 35°$.
Angle $PRQ$ is obtuse.



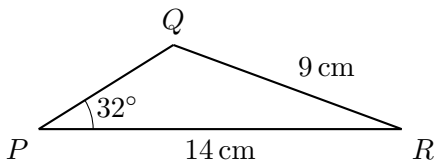

**(a)** Calculate angle $PRQ$.

Angle $PRQ =$ ..... ° **[3]**

**(b)** Calculate the area of triangle $PQR$.

Area = ..... $cm^2$ **[3]**

**15** The diagram shows triangle $PQR$.
The angle $PQR$ is obtuse.



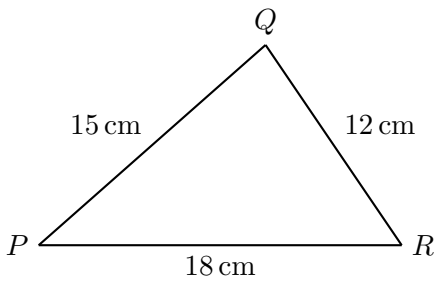

**(a)** Calculate the obtuse angle $PQR$.

..... ° **[4]**

**(b)** Calculate the area of triangle $PQR$.

..... $cm^2$ **[3]**

**16** The diagram shows triangle $PQR$.
$PQ = 15\,\text{cm}$, $QR = 12\,\text{cm}$ and $PR = 18\,\text{cm}$.




**(a)** Use the cosine rule to show that angle $PQR = 82.8°$, correct to 1 decimal place.

**[3]**

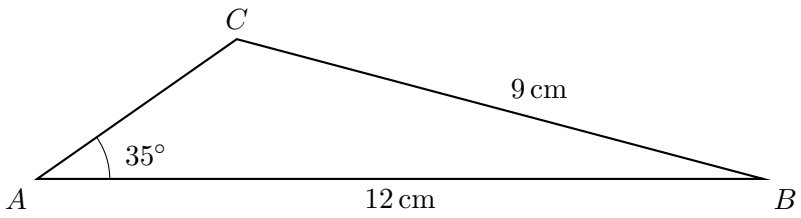
**(b)** Calculate the area of triangle $PQR$.

..... $cm^2$ **[2]**

**(c)** Calculate the shortest distance from $R$ to the line $PQ$.

..... cm **[2]**

**17** In triangle $ABC$, $AB = 12\,\text{cm}$, $BC = 9\,\text{cm}$ and angle $BAC = 35°$.




NOT TO SCALE

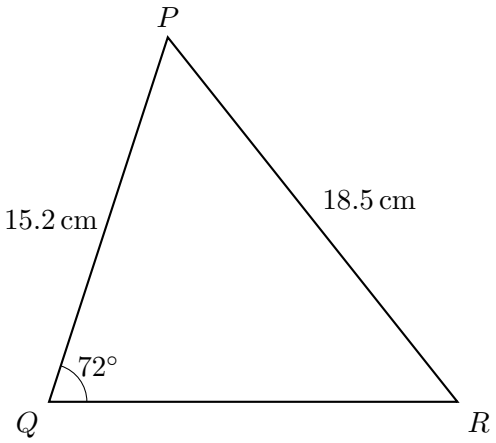
**(a)** Calculate the two possible values of angle $ACB$.
Give your answers correct to 1 decimal place.

..... ° **[4]**

**(b)** Given that angle $ACB$ is obtuse, calculate the area of triangle $ABC$.

Area = ..... $cm^2$ **[3]**

**18** The diagram shows a triangle $PQR$.
$PQ = 15.2\,\text{cm}$, $PR = 18.5\,\text{cm}$ and angle $PQR = 72°$.




NOT TO SCALE

**(a)** Calculate the size of angle $PRQ$.

Angle $PRQ =$ ..... ° **[3]**

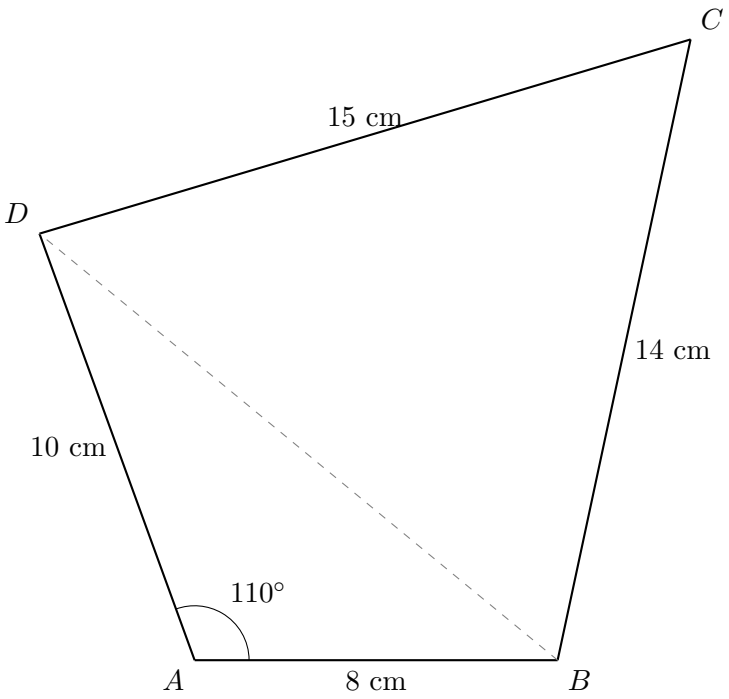
**(b)** Calculate the length of $QR$.

$QR =$ ..... cm **[3]**

**(c)** Calculate the shortest distance from $P$ to the line $QR$.

Distance = ..... cm **[2]**

**19** The diagram shows a quadrilateral $ABCD$.
$AB = 8$ cm, $AD = 10$ cm, $BC = 14$ cm and $CD = 15$ cm.
Angle $BAD = 110°$.




**(a)** Calculate the length of the diagonal $BD$.

..... cm **[3]**

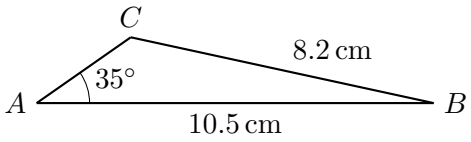
**(b)** Calculate the angle $BCD$.

..... ° **[3]**

**(c)** Calculate the area of the quadrilateral $ABCD$.

..... $cm^2$ **[3]**

**20** In triangle $ABC$, $AB = 10.5\,\text{cm}$, $BC = 8.2\,\text{cm}$ and angle $BAC = 35°$.
Angle $BCA$ is obtuse.




**(a)** Calculate the size of angle $BCA$.

..... ° **[3]**

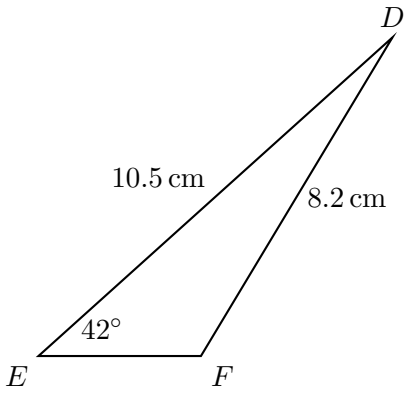
**(b)** Calculate the length of $AC$.

$AC =$ ..... cm **[3]**

**(c)** A point $D$ lies outside the triangle $ABC$ such that $CD = 6.4\,\text{cm}$ and angle $ACD$ is acute.
The area of triangle $ACD$ is $8.5\,\text{cm}^2$.
Calculate the size of angle $ACD$.

..... ° **[2]**

**21** The diagram shows triangle $DEF$.
$DE = 10.5$ cm, $DF = 8.2$ cm and angle $DEF = 42°$.
Angle $DFE$ is an obtuse angle.




**(a)** Calculate angle $DFE$.

Angle $DFE =$ ..... ° **[3]**

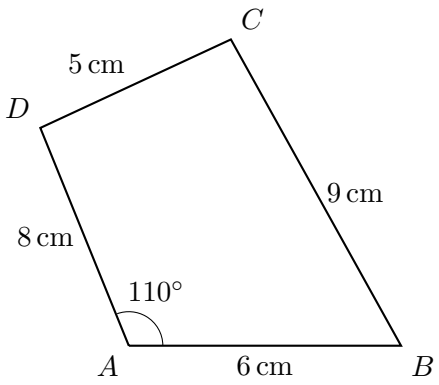
**(b)** Calculate the length of $EF$.

$EF =$ ..... cm **[3]**

**(c)** Calculate the shortest distance from $D$ to the line $EF$.

..... cm **[2]**

**22** The diagram shows a quadrilateral $ABCD$.
$AB = 6$ cm, $AD = 8$ cm, $DC = 5$ cm, and $BC = 9$ cm.
Angle $DAB = 110°$.




NOT TO SCALE

**(a)** Calculate the length of the diagonal $BD$.

..... *cm* **[3]**

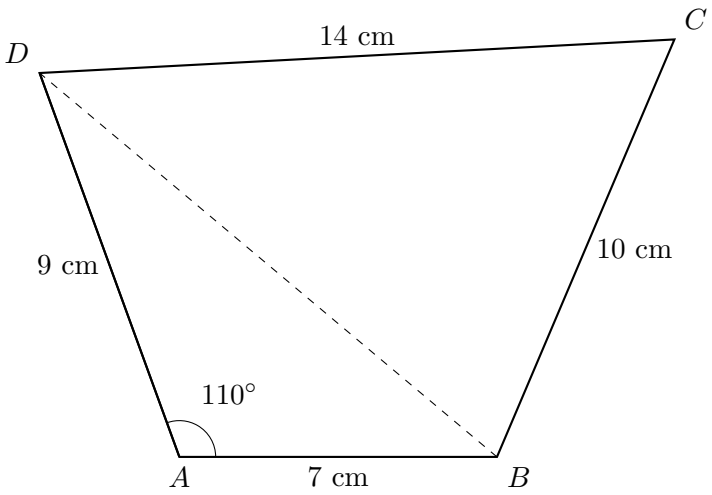
**(b)** Calculate the size of angle $BCD$.

..... ° **[3]**

**(c)** Calculate the total area of the quadrilateral $ABCD$.

..... $cm^2$ **[3]**

**23** The diagram shows a quadrilateral $ABCD$.
$AB = 7$ cm, $AD = 9$ cm, $BC = 10$ cm and $CD = 14$ cm.
Angle $BAD = 110°$.




**(a)** Calculate the length of $BD$.

..... cm **[3]**

**(b)** Calculate angle $BCD$.

..... ° **[3]**

**(c)** Calculate the total area of the quadrilateral $ABCD$.

..... $cm^2$ **[3]**

**24** The diagram shows a quadrilateral $ABCD$.
$AB = 7.4$ cm, $BC = 9.1$ cm, and $CD = 10.5$ cm.
Angle $ABC = 112°$ and angle $CAD = 43°$.

**(a)** Calculate the length of $AC$.

..... cm **[3]**

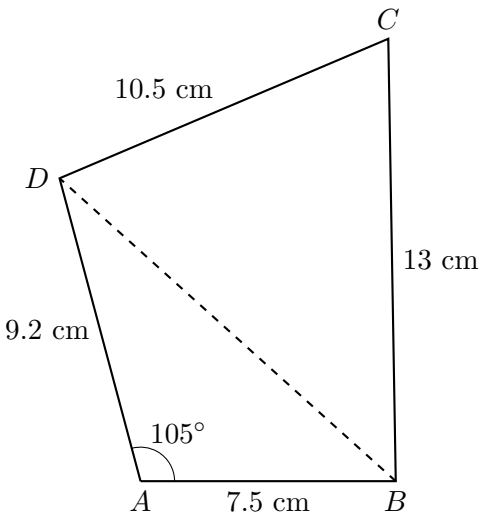
**(b)** Calculate the acute angle $ADC$.

..... ° **[3]**

**(c)** Calculate the shortest distance from $D$ to the line $AC$.

..... cm **[3]**

**25** The diagram shows a quadrilateral $ABCD$.
$AB = 7.5$ cm, $AD = 9.2$ cm and angle $BAD = 105°$.
$BC = 13$ cm and $CD = 10.5$ cm.




**(a)** Calculate the length of $BD$.

$BD =$ ..... cm **[3]**

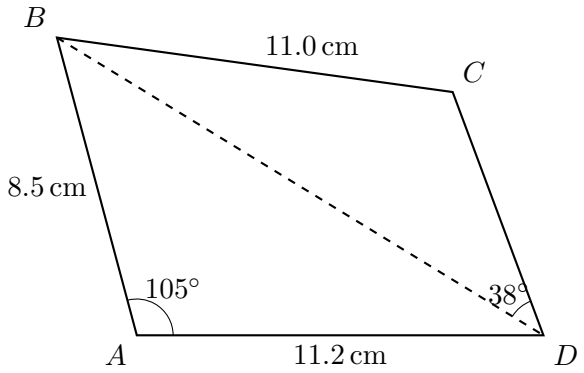
**(b)** Calculate angle $BCD$.

Angle $BCD =$ ..... ° **[3]**

**(c)** Calculate the area of the quadrilateral $ABCD$.

Area = ..... $cm^2$ **[3]**

**26** The diagram shows a quadrilateral $ABCD$.
$AB = 8.5\,\text{cm}$, $AD = 11.2\,\text{cm}$ and angle $BAD = 105°$.
$BC = 11.0\,\text{cm}$ and angle $BDC = 38°$.
Angle $BCD$ is obtuse.




NOT TO SCALE

**(a)** Calculate the length of $BD$.

$BD =$ ..... cm **[3]**

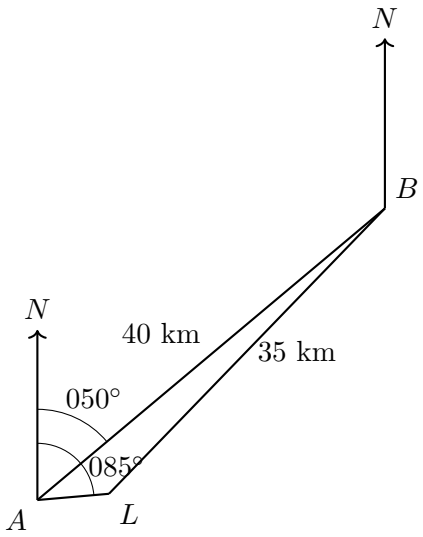
**(b)** Calculate the size of angle $BCD$.

Angle $BCD =$ ..... ° **[3]**

**(c)** Calculate the area of the quadrilateral $ABCD$.

Area = ..... $cm^2$ **[4]**

**27** Ship $A$ and Ship $B$ are at sea.
Ship $B$ is 40 km from Ship $A$ on a bearing of 050°.
A lighthouse $L$ is 35 km from Ship $B$.
The bearing of $L$ from $A$ is 085°.
Given that the angle $ALB$ is obtuse:




**(a)** Calculate the obtuse angle $ALB$.

..... ° **[4]**

**(b)** Calculate the distance $AL$.

..... km **[3]**

**(c)** Calculate the bearing of $L$ from $B$.

..... ° **[3]**

**1** **(a)** $7^2 = 5^2 + x^2 - 2(5)(x)\cos 60°$ **[M1]**

$\cos 60° = \frac{1}{2}$ or 0.5 seen **[B1]**

$49 = 25 + x^2 - 5x \implies x^2 - 5x - 24 = 0$ (with no errors seen) **[A1]**

**(b)** $(x - 8)(x + 3) = 0$ **[M1]**

$x = 8$ **[A1]**

**(c)** Area $= \frac{1}{2} \times 5 \times 8 \times \sin 60°$ **[M1]**

$10\sqrt{3}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

**2** **(a)** $0.5 \times 8 \times 12 \times \sin(42°)$ **[M1]**

32.1 (or 32.11 to 32.12) **[A1]**

**(b)** $BC^2 = 8^2 + 12^2 - 2(8)(12)\cos(42°)$ **[M1]**

$BC^2 = 65.3...$ **[M1]**

8.08 (or 8.081 to 8.082) **[A1]**

[Back to Question] *[Total Marks: 5]*

**3** **(a)** $\frac{1}{2} \times 8 \times 12 \times \sin 40°$ **[M1]**

30.9 (accept 30.85 . . . ) **[A1]**

**(b)** $8^2 + 12^2 - 2 \times 8 \times 12 \times \cos 40°$ **[M1]**

60.9 . . . **[A1]**

7.80 or 7.81 **[A1]**

[Back to Question] *[Total Marks: 5]*

**4** **(a)** $0.5 \times 8.5 \times 11.2 \times \sin(54°)$ **[M1]**

38.5 (38.50 to 38.51) **[A1]**

**(b)** $BC^2 = 8.5^2 + 11.2^2 - 2(8.5)(11.2)\cos(54°)$ [M2]
(or [M1] for correct cosine rule formula with one substitution error)

$BC^2 = 85.77...$ **[A1]**

9.26 (9.261...) **[A1]**

[Back to Question] *[Total Marks: 5]*

**5** **(a)** $\frac{1}{2} \times 8 \times 12 \times \sin(50°)$ **[M1]**

36.77... so 36.8 **[A1]**

**(b)** $BC^2 = 8^2 + 12^2 - 2(8)(12)\cos(50°)$ **[M1]**

$BC^2 = 208 - 123.415... = 84.58...$ **[M1]**

9.197... so 9.20 **[A1]**

[Back to Question] *[Total Marks: 5]*

**6 (a)** $\frac{1}{2} \times 7.2 \times 9.5 \times \sin 64°$ **[M1]**

30.7 or 30.73... **[A1]**

**(b)** $7.2^2 + 9.5^2 - 2(7.2)(9.5)\cos 64°$ **[M2]**

(or $M1$ for correct implicit cosine rule equation)

9.06 or 9.062... **[A1]**

[Back to Question] *[Total Marks: 5]*

**7 (a)** Uses cosine rule: $BC^2 = 8^2 + 12^2 - 2(8)(12)\cos 65°$ **[M1]**

$BC^2 = 126.85...$ **[M1]**

11.3 **[A1]**

**(b)** Uses area formula: $\frac{1}{2} \times 8 \times 12 \times \sin 65°$ **[M1]**

43.5 **[A1]**

[Back to Question] *[Total Marks: 5]*

**8 (a)** $\frac{1}{2} \times 8 \times 11 \times \sin(65°)$ **[M1]**

39.9 **[A1]**

**(b)** $BC^2 = 8^2 + 11^2 - 2(8)(11)\cos(65°)$ **[M2]**

($M1$ for correct substitution into implicit cosine rule)

10.5 **[A1]**

[Back to Question] *[Total Marks: 5]*

**9 (a)** $\frac{BC}{\sin 45°} = \frac{12}{\sin 60°}$ **[M1]**

$BC = \frac{12 \sin 45°}{\sin 60°}$ **[M1]**

9.80 **[A1]**

**(b)** Angle $ACB = 180° - (45° + 60°) = 75°$ **[B1]**

Area $= \frac{1}{2} \times 12 \times 9.798 \times \sin 75°$ **[M1]**

56.8 **[A1]**

[Back to Question] *[Total Marks: 6]*

**10** **(a)** $\dfrac{BC}{\sin 40°} = \dfrac{12}{\sin 60°}$ **[M1]**

$BC = \dfrac{12 \sin 40°}{\sin 60°}$ **[M1]**

8.91 (accept 8.907 . . . ) **[A1]**

**(b)** Angle $BCA = 180° - (40° + 60°) = 80°$ **[B1]**

$\dfrac{1}{2} \times 12 \times 8.91 \times \sin 80°$ **[M1]**

52.6 (accept 52.62 . . . to 52.65 . . . ) **[A1]**

[Back to Question] *[Total Marks: 6]*

**11** **(a)** $\cos Q = \dfrac{7^2 + 8^2 - 12^2}{2 \times 7 \times 8}$ [M2] (M1 for correct implicit cosine rule equation e.g. $12^2 = 7^2 + 8^2 - 2(7)(8)\cos Q$)

106.1 (accept 106.07 . . . ) **[A1]**

**(b)** $\dfrac{1}{2} \times 7 \times 8 \times \sin 106.1°$ **[M1]**

26.9 (accept 26.89 . . . ) **[A1]**

[Back to Question] *[Total Marks: 5]*

**12** **(a)** $\dfrac{\sin P}{12} = \dfrac{\sin 35°}{8}$ **[M1]**

$\sin P = \dfrac{12 \sin 35°}{8} = 0.8603...$ **[M1]**

Acute angle $= 59.35...°$ **[A1]**

Obtuse angle $P = 180° - 59.35...° = 120.64...°$ so $120.6°$ **[A1]**

**(b)** Angle $R = 180° - 35° - 120.64...° = 24.36...°$ **[M1]**

Area $= \dfrac{1}{2} \times 8 \times 12 \times \sin(24.36...°) = 19.79...$ so 19.8 **[A1]**

[Back to Question] *[Total Marks: 6]*

**13** **(a)** $\dfrac{\sin R}{15} = \dfrac{\sin 42°}{12}$ **[M1]**

$\sin R = \dfrac{15 \sin 42°}{12}$ (evaluates to 0.836 . . . ) **[M1]**

123.2 (accept 123.2 to 123.3) **[A1]**

**(b)** Angle $P = 180° - 42° - 123.24° = 14.76° \ldots$ **[M1]**

$\dfrac{QR}{\sin 14.76°} = \dfrac{12}{\sin 42°}$ (or equivalent application of sine/cosine rule) **[M1]**

4.57 (accept 4.56 to 4.58) **[A1]**

[Back to Question] *[Total Marks: 6]*

**14** **(a)** $\dfrac{\sin R}{14} = \dfrac{\sin 35°}{10}$ **[M1]**

$\sin^{-1}\left(\dfrac{14\sin 35°}{10}\right)$ leading to $53.4°$ $(53.41...)$ **[M1]**

$126.6$ $(126.58...)$ **[A1]**

**(b)** Angle $PQR = 180° - 35° - 126.58° = 18.4°$ **[M1]**

Area $= \dfrac{1}{2} \times 14 \times 10 \times \sin(18.4°)$ **[M1]**

$22.1$ $(22.11...)$ **[A1]**

**[Back to Question]** *[Total Marks: 6]*

**15** **(a)** $\dfrac{\sin Q}{14} = \dfrac{\sin 32°}{9}$ **[M1]**

$\sin Q = \dfrac{14\sin 32°}{9}$ or $0.824...$ **[M1]**

Acute angle $= 55.5°$ $(55.51...)$ **[A1]**

$180 - 55.5... = 124.5°$ $(124.48$ to $124.5)$ **[A1]**

**(b)** Angle $PRQ = 180 - 32 - 124.48 = 23.52°$ **[B1]**

$0.5 \times 14 \times 9 \times \sin(23.52°)$ **[M1]**

$25.1$ $(25.12$ to $25.14)$ **[A1]**

**[Back to Question]** *[Total Marks: 7]*

**16** **(a)** $\cos Q = \dfrac{15^2 + 12^2 - 18^2}{2 \times 15 \times 12}$ **[M1]**

$\cos Q = \dfrac{45}{360}$ OR $0.125$ **[M1]**

$Q = \cos^{-1}(0.125) = 82.819...°$ **[A1]**

**(b)** $0.5 \times 15 \times 12 \times \sin(82.8°)$ **[M1]**

$89.3$ (or $89.28$ to $89.30$) **[A1]**

**(c)** Area $= 0.5 \times 15 \times h = 89.3$ OR $h = 12 \times \sin(82.8°)$ **[M1]**

$11.9$ (or $11.90$ to $11.91$) **[A1]**

**[Back to Question]** *[Total Marks: 7]*

**17** **(a)** $\dfrac{\sin C}{12} = \dfrac{\sin 35°}{9}$ **[M1]**

$\sin C = \dfrac{12\sin 35°}{9}$ $(= 0.764...)$ **[M1]**

$49.9°$ (or $49.88...$) **[A1]**

$130.1°$ (or $130.11...$) **[A1]**

**(b)** Angle $ABC = 180 - (35 + 130.1) = 14.9°$ **[M1]**

Area $= \frac{1}{2} \times 12 \times 9 \times \sin(14.9°)$ **[M1]**

13.9 (accept 13.87... to 13.89...) **[A1]**

[Back to Question] *[Total Marks: 7]*

**18** **(a)** Uses sine rule: $\frac{\sin R}{15.2} = \frac{\sin 72°}{18.5}$ **[M1]**

$\sin R = 0.781...$ **[M1]**

51.4 **[A1]**

**(b)** Angle $P = 180° - 72° - 51.38° = 56.62°$ **[B1]**

$\frac{QR}{\sin 56.62°} = \frac{18.5}{\sin 72°}$ **[M1]**

16.2 **[A1]**

**(c)** Shortest distance $= 15.2 \times \sin 72°$ **[M1]**

14.5 **[A1]**

[Back to Question] *[Total Marks: 8]*

**19** **(a)** $BD^2 = 8^2 + 10^2 - 2(8)(10)\cos 110°$ **[M1]**

$BD^2 = 164 - 160(-0.342) = 218.72$ **[M1]**

14.8 **[A1]**

**(b)** $\cos C = \frac{14^2 + 15^2 - 14.79^2}{2(14)(15)}$ **[M1]**

$\cos C = \frac{202.26}{420} = 0.4816$ **[M1]**

61.2 **[A1]**

**(c)** Area $ABD = \frac{1}{2}(8)(10)\sin 110° = 37.59$ **[M1]**

Area $BCD = \frac{1}{2}(14)(15)\sin 61.2° = 92.01$ **[M1]**

130 (or 129.6) **[A1]**

[Back to Question] *[Total Marks: 9]*

**20** **(a)** $\frac{\sin C}{10.5} = \frac{\sin 35°}{8.2}$ **[M1]**

$\sin C = \frac{10.5 \times \sin 35°}{8.2}$ OR 0.734... **[M1]**

$C = 180° - 47.3° = 132.7°$ (or 132.73 to 132.74) **[A1]**

**(b)** Angle $B = 180° - 35° - 132.7° = 12.3°$ **[M1]**

$\frac{AC}{\sin 12.3°} = \frac{8.2}{\sin 35°}$ OR $AC^2 = 10.5^2 + 8.2^2 - 2(10.5)(8.2)\cos 12.3°$ **[M1]**

3.04 (or 3.035 to 3.045) **[A1]**

**(c)** $0.5 \times 3.04 \times 6.4 \times \sin(ACD) = 8.5$ **[M1]**

$\sin(ACD) = \dfrac{8.5}{9.728}$ OR Angle $ACD = \sin^{-1}\left(\dfrac{8.5}{9.728}\right) = 60.9°$ (or 60.8... to 61.1°) **[A1]**

[Back to Question] *[Total Marks: 8]*

**21** **(a)** $\dfrac{\sin F}{10.5} = \dfrac{\sin 42°}{8.2}$ **[M1]**

$\sin F = 0.8568\ldots$ yielding acute angle $58.96\ldots°$ **[A1]**

$180° - 58.96° = 121.0$ (accept $121.04\ldots$) **[A1]**

**(b)** Angle $EDF = 180° - 42° - 121.04° = 16.96°$ **[B1]**

$\dfrac{EF}{\sin 16.96°} = \dfrac{8.2}{\sin 42°}$ **[M1]**

3.57 (accept $3.56\ldots$ to $3.58\ldots$) **[A1]**

**(c)** Shortest distance $= 10.5 \times \sin 42°$ **[M1]**

7.03 (accept $7.025\ldots$ to $7.027\ldots$) **[A1]**

[Back to Question] *[Total Marks: 8]*

**22** **(a)** $BD^2 = 6^2 + 8^2 - 2(6)(8)\cos(110°)$ **[M1]**

$BD^2 = 100 - 96(-0.342...) = 132.83...$ **[M1]**

$BD = 11.52...$ so 11.5 **[A1]**

**(b)** $\cos C = \dfrac{9^2 + 5^2 - (11.525...)^2}{2(9)(5)}$ **[M1]**

$\cos C = \dfrac{106 - 132.83...}{90} = -0.2981...$ **[M1]**

$C = 107.34...°$ so 107.3° **[A1]**

**(c)** Area of triangle $ABD = \dfrac{1}{2} \times 6 \times 8 \times \sin(110°) = 22.55...$ **[M1]**

Area of triangle $BCD = \dfrac{1}{2} \times 9 \times 5 \times \sin(107.34...°) = 21.47...$ **[M1]**

Total Area $= 22.55... + 21.47... = 44.02...$ so 44.0 **[A1]**

[Back to Question] *[Total Marks: 9]*

**23** **(a)** $7^2 + 9^2 - 2 \times 7 \times 9 \times \cos 110°$ **[M1]**

$173.09\ldots$ **[A1]**

13.2 (accept 13.15 to 13.2) **[A1]**

**(b)** $13.156^2 = 10^2 + 14^2 - 2 \times 10 \times 14 \times \cos(\angle BCD)$ **[M1]**

$\cos(\angle BCD) = \dfrac{10^2 + 14^2 - 173.09}{2 \times 10 \times 14}$ **[M1]**

64.0 (accept $63.9\ldots$ to 64.0) **[A1]**

**(c)** $\frac{1}{2} \times 7 \times 9 \times \sin 110°$ (evaluates to 29.6 . . . ) **[M1]**

$\frac{1}{2} \times 10 \times 14 \times \sin 64.0°$ (evaluates to 62.9 . . . ) **[M1]**

92.5 (accept 92.4 to 92.6) **[A1]**

**[Back to Question]** *[Total Marks: 9]*

**24 (a)** $AC^2 = 7.4^2 + 9.1^2 - 2(7.4)(9.1)\cos(112°)$ [M2]
(or [M1] for correct cosine rule formula with one error)

$AC = 13.7$ (13.71 to 13.72) **[A1]**

**(b)** $\frac{\sin D}{13.712} = \frac{\sin 43°}{10.5}$ **[M1]**

$\sin D = \frac{13.712 \sin 43°}{10.5}$ or 0.890... **[M1]**

63.0 (62.95 to 63.0) **[A1]**

**(c)** Angle $ACD = 180 - 43 - 62.95 = 74.05°$ **[M1]**

Shortest distance $= 10.5 \times \sin(74.05°)$ **[M1]**

10.1 (10.09 to 10.1) **[A1]**

**[Back to Question]** *[Total Marks: 9]*

**25 (a)** $BD^2 = 7.5^2 + 9.2^2 - 2(7.5)(9.2)\cos(105°)$ **[M2]**

($M1$ for correct substitution into implicit cosine rule)

13.3 (13.28...) **[A1]**

**(b)** $\cos C = \frac{13^2 + 10.5^2 - 13.289^2}{2(13)(10.5)}$ **[M2]**

($M1$ for correct substitution into implicit cosine rule)

67.9 (67.91...) **[A1]**

**(c)** Area $ABD = \frac{1}{2} \times 7.5 \times 9.2 \times \sin(105°)$ (33.32...) **[M1]**

Area $BCD = \frac{1}{2} \times 13 \times 10.5 \times \sin(67.9°)$ (63.24...) **[M1]**

96.6 (96.5... to 96.6) **[A1]**

**[Back to Question]** *[Total Marks: 9]*

**26 (a)** Uses cosine rule: $BD^2 = 8.5^2 + 11.2^2 - 2(8.5)(11.2)\cos 105°$ **[M1]**

$BD^2 = 246.96...$ **[A1]**

15.7 **[A1]**

**(b)** Uses sine rule: $\frac{\sin C}{15.715} = \frac{\sin 38°}{11.0}$ **[M1]**

Acute angle $C = 61.59...°$ **[M1]**

Obtuse angle $BCD = 180° - 61.59...° = 118.4°$ **[A1]**

**(c)** Area of triangle $ABD = \frac{1}{2} \times 8.5 \times 11.2 \times \sin 105°$ $(= 45.97...)$ **[M1]**

Angle $DBC = 180° - 38° - 118.41° = 23.59°$ **[B1]**

Area of triangle $BCD = \frac{1}{2} \times 11.0 \times 15.715 \times \sin 23.59°$ $(= 34.59...)$ **[M1]**

80.6 **[A1]**

**[Back to Question]** *[Total Marks: 10]*

**27** **(a)** Angle $BAL = 085° - 050° = 35°$ **[B1]**

$\frac{\sin(ALB)}{40} = \frac{\sin 35°}{35}$ **[M1]**

$\sin(ALB) = \frac{40 \sin 35°}{35} = 0.6555\ldots$ **[M1]**

Acute angle $= 40.96°$, Obtuse angle $ALB = 180° - 40.96° = 139.0$ **[A1]**

**(b)** Angle $ABL = 180° - 35° - 139.04° = 5.96°$ **[B1]**

$\frac{AL}{\sin 5.96°} = \frac{35}{\sin 35°}$ **[M1]**

6.33 **[A1]**

**(c)** Bearing of $A$ from $B = 180° + 050° = 230°$ **[M1]**

Bearing of $L$ from $B = 230°$ − Angle $ABL = 230° - 5.96°$ **[M1]**

224.0 **[A1]**

**[Back to Question]** *[Total Marks: 10]*